PROGRESS REPORT

Bill Scanning Reward & Intelligence System

An OCR + ML pipeline for receipt scanning, fraud detection and intelligent reward allocation

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1,440	100	5	0.94
training samples	Indian receipts	fraud signals	baseline F1

This report summarises data collection, exploratory analysis, fraud-detection design and preliminary modelling results for the current sprint.

1. Introduction & Methodology

The objective of this project is to build an end-to-end pipeline that scans receipts, extracts structured data using OCR, and detects fraudulent manipulations before reward points are issued. The dataset comprises three primary sources, consolidated into a single training corpus.

CORD
Public dataset - annotated receipt images

SROIE
Public dataset - scanned receipt images

Indian Receipts
100 manual photos - shops, food, pharmacy

Data Provenance Map

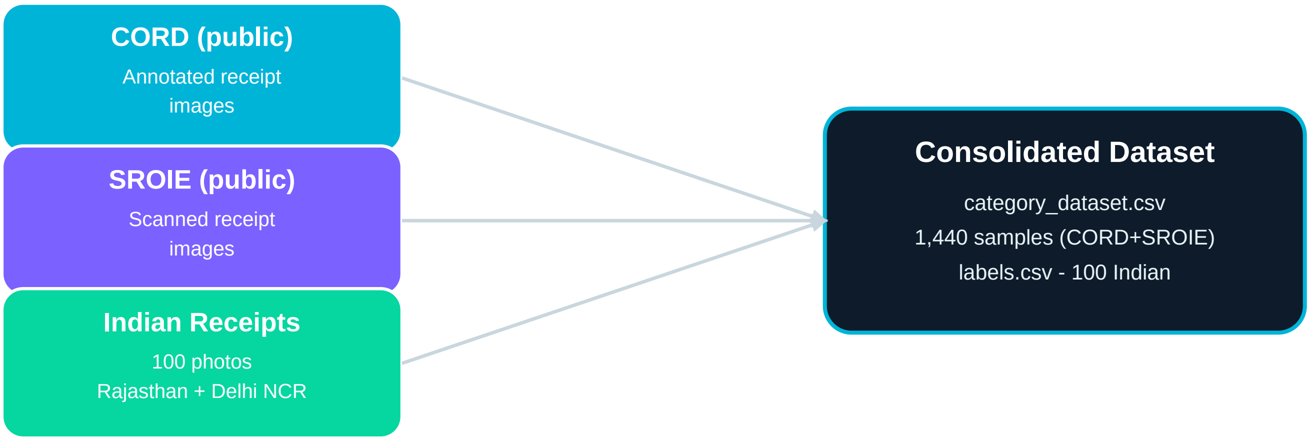

Figure 1: Sources consolidated into the training corpus. Indian receipts collected across Rajasthan and Delhi NCR.

Labelling Process (Jyoti)

The 100 Indian receipts were manually labelled into four categories:

genuine
Clean, valid receipt, no tampering

tampered
Edits, bad totals, duplicated content

multi_bill
One image with multiple bills

handwritten_edit
Handwritten edits on printed bill

Tampered Generation (Ranjeet): Using `generate_tampered.py`, additional tampered versions (v1 and v2) of the Indian receipts were created via brightness adjustment, number overwriting and duplicate copying - enriching the fraud-detection training set.

System Pipeline: Bill → Reward

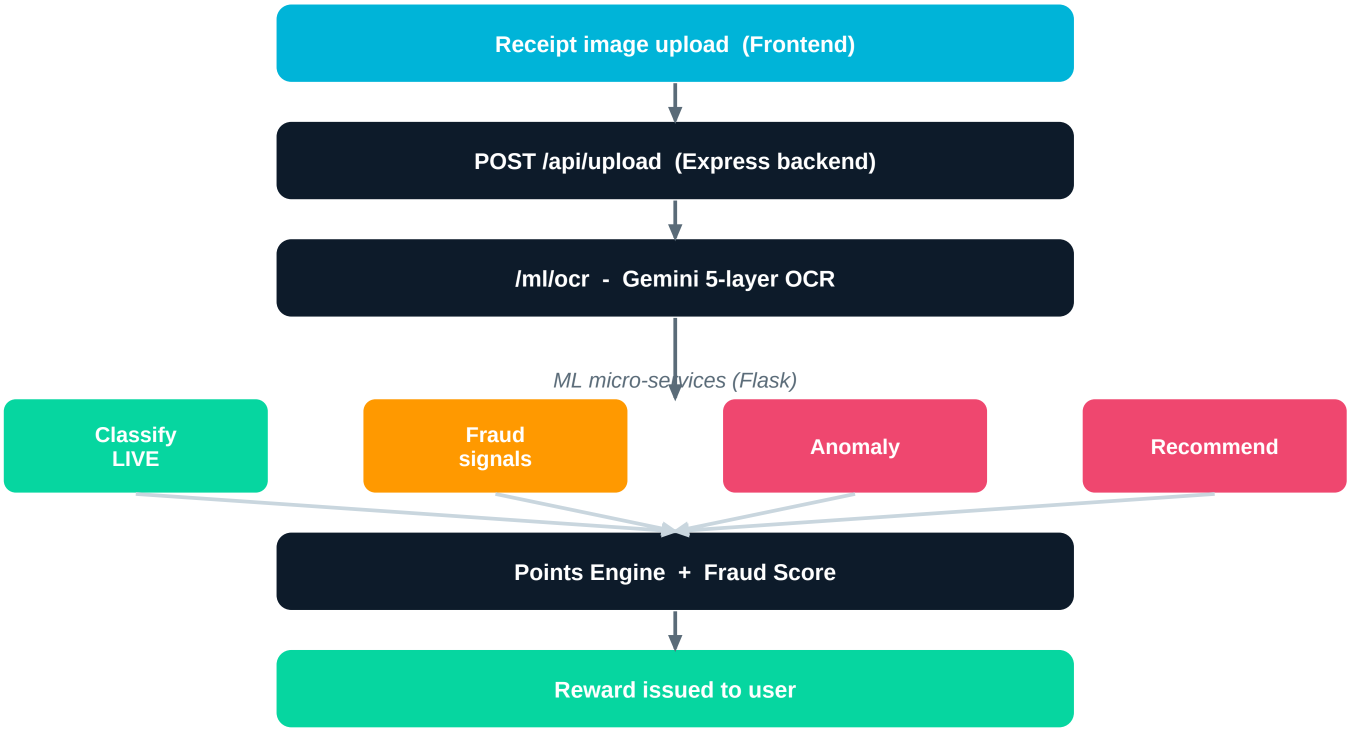

Figure 2: End-to-end flow. Green = live, orange = partial, red = in progress.

2. Dataset Statistics & Exploratory Analysis

2.1 Overall Dataset Composition

The combined dataset (`category_dataset.csv`) contains **1,440 samples** from CORD and SROIE, distributed across two broad categories:

Category	Count	Share
Restaurant	460	31.9%
Retail	980	68.1%
Total	1,440	100%

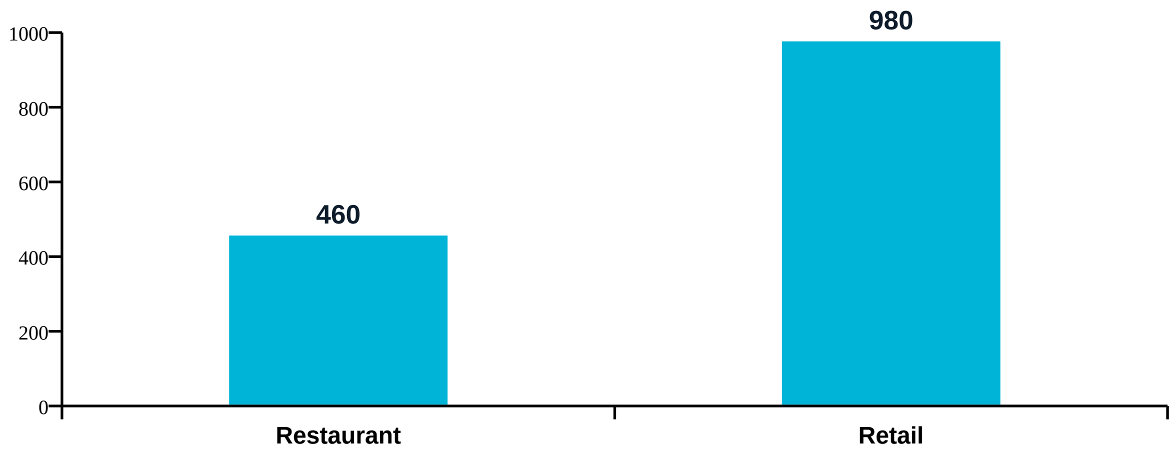

Figure 3: Category distribution across the combined corpus.

2.2 Null Value Analysis

A thorough null check revealed missing values across metadata columns. This is expected in raw OCR output and is handled by the pipeline's fallback logic.

Column	Missing Count
merchant	454
address	455
date	608
total	3
items_text	973

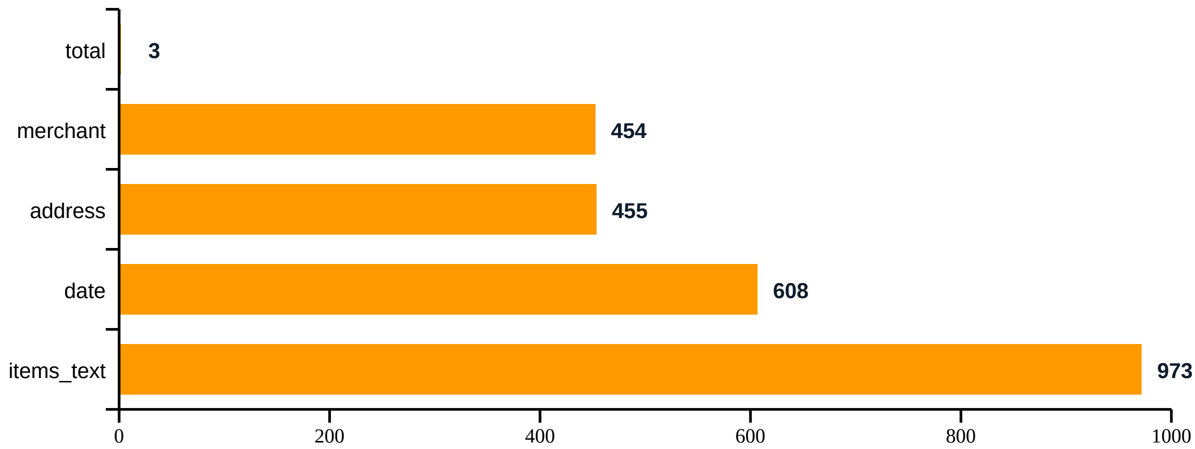

Figure 4: Missing values by column (out of 1,440 rows).

2.3 Fraud Label Distribution (100 Indian Receipts)

The labelled dataset (`labels.csv`) shows a healthy split for training a fraud classifier - both genuine and tampered classes are well represented.

Label	Percentage
Genuine	64.5%
Tampered	32.5%
Multi-bill	2.5%
Handwritten	0.5%

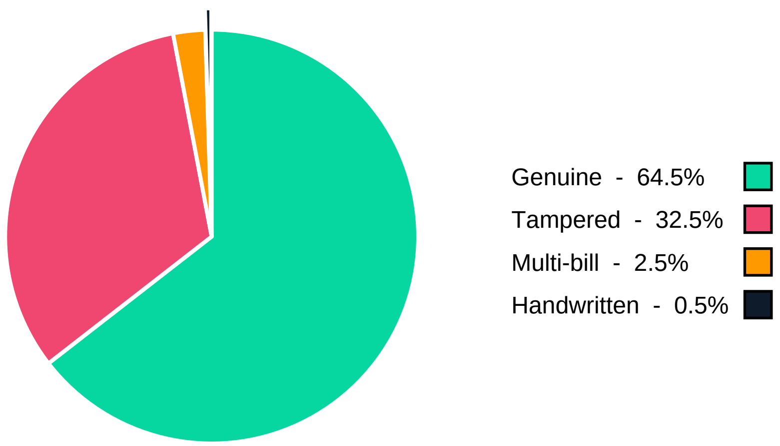

Figure 5: Fraud-label distribution across the 100 labelled Indian receipts.

3. Fraud Detection Signals & OCR Challenges

Fraud detection is not limited to manual labels. Automated challenge handlers flag suspicious receipts; these signals aggregate into a final fraud score per upload.

#	Signal	Technique
1	Blur detection	OpenCV Laplacian variance; rejects images below threshold (100).
2	Multi-bill anomaly	Flags abnormally high text-region counts (merged documents).
3	Duplicate check	Perceptual hash (pHash) + merchant-date-total matching.
4	Handwritten edits	Gemini confidence + pixel-density analysis of overwrites.
5	Rate-limit handling	Exponential backoff for external API calls (Gemini).

4. Preliminary Model Performance

A classifier categorises receipts into **Restaurant vs Retail** (feeding the reward rules). The first baseline (Random Forest) has been evaluated on the validation set.

Model	Macro F1-Score	Status
Random Forest (baseline)	0.94	Evaluated
XGBoost	-	Training
Logistic Regression	-	Training

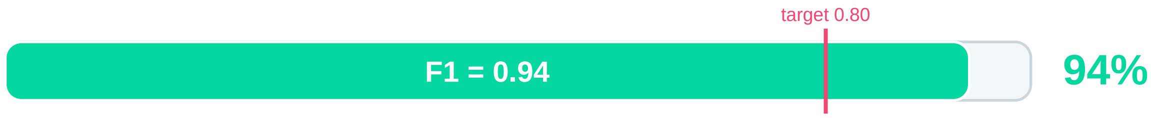

Figure 6: Baseline macro-F1 (0.94) comfortably clears the 0.80 target. XGBoost and Logistic Regression will be added once finalised.

5. Next Steps & Conclusion

The team is on track for the Day 12 deadline. Immediate next steps:

#	Action	Owner
1	Integrate all OCR challenge handlers with the Flask backend	Ashfaaq
2	Merge fraud_detector.py with the server upload route	Ashfaaq / Ranjeet
3	End-to-end test on 50 sample receipts (record TP/FP rates)	Ranjeet
4	Finalise live demo (clean / blurry / tampered receipts)	Team

Conclusion. The dataset is documented, the baseline model is showing promising results (F1 = 0.94), and the team is aligned for the final integration phase. We are confident of delivering a working end-to-end demo for the review.

Team ARAJ

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